

JAMES GRAY

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EDUCATION

University of Warwick – BEng Computer Systems Engineering (Year in Industry)

Sept. 2022 - June 2026

Grade: First (83.3%)

Key Modules: AI, Data Structures, Operating Systems & Networks, Data Analytics, Computer Architecture, FPGAs, Software Engineering, Maths

Reading School

Sept. 2020 - July 2022

Grades: A*A*A* (Maths, Physics, Computer Science)

SKILLS & AWARDS

Skills: C, C++, C#, Python, Java, SQL, React, Typescript, HTML, CSS, JS, Haskell, Matlab, Verilog

Awards: Exceptional Achievement in Second Year, Computer Science Department Award

WORK EXPERIENCE

Qube Research and Technologies – Quantitive Technology Intern

June 2024 – Present

- Improving deployments by developing a tool that automatically generates environment-specific configurations, new supporting tools that leverage these configurations, and integrating the process management service to enhance consistency, reduce manual effort, and establish an explicit single source of truth.
- Designed and implemented a C++ process management service that enables support teams to control and monitor system processes through both a UI and code, eliminating the need for manual SSH interaction and providing hands-on experience with Unix system programming and service design.
- Developed automated tools to generate documentation and samples for multiple languages to a single reference point.
- Reworked and upgraded system components and the CI/CD pipeline to support C# builds and tests on the Linux environment, increasing code coverage and Linux developer capabilities.
- Developed an LLM-based support assistant with a React.js frontend to generate code samples and deliver system-specific responses which used the Reason & Act framework with custom tools to allow multi-step reasoning, domain-specific context access, and output verification to significantly improve response accuracy.

MyTutor – Tutor

Jan. 2023 – Present

- Communicated complex concepts in a simple way to help A-Level and GCSE students master subject material.

Expedia – Spring Insight

April 2022 – April 2022

- Gained practical insights into the processes and agile methodologies employed in software engineering and mobile app development projects.

Cisco – Software Engineering Course

June 2019 – June 2019

- Gained practical insight into the software development process by building and iterating on basic Python programs, and familiarised with Git for collaboration.

Oracle – Work Experience

May 2019 – May 2019

- Developed a basic web-based chatbot with a Java Spring backend and a HTML, CSS and JS frontend, gaining hands-on experience with RESTful APIs and using Gradle for dependency management and project build automation.

KEY PROJECTS

FPGA Pacman

(Verilog)

A recreation of Pacman on a Nexys 4 FPGA using Verilog; achieved the top score in the year.

- Debugged hardware signals using Verilog test benches in Vivado to verify signals and timings across different digital modules.
- Implemented custom VGA drivers to manage the graphics output, including frame buffer control, pixel timing generation.

Magnetic Electron Trap Simulation

(Python, Scipy, Numpy, Matplotlib, Optimisation)

Solved and visualised differential equations using Matplotlib, SciPy and NumPy to model an electron within a magnetic field; achieved a score of 100%.

- Implemented multiprocessing to parallelise simulation code and bypass the Global Interpreter Lock (GIL) to achieve a 20x performance improvement.

Multithreaded Packet Sniffer

(C, Networking, Multithreading)

Developed a multithreaded packet sniffer to identify domain blacklist violations, SYN attacks, and ARP cache poisoning attacks.

- Implemented a thread-safe packet queue with pthreads to dispatch work efficiently to a thread pool, allowing the application to successfully process 1,000,000s of packets without loss.
- Ensured memory leak, race condition and bug free code by validating code with tools like Valgrind, Helgrind and GDB.

Optimising the Conjugate Gradient Numerical Method on a 3D Mesh

(C, Vector Instructions, Cache and Memory Optimisation)

- Optimised the conjugate gradient numerical method on a 3D mesh by leveraging AVX-256 intrinsics, OpenMP directives, and code refactoring to optimise memory data locality, to achieve an 8.88x speedup.

Movie Information Viewer

(Java, Data Structures)

- Implemented data structures and abstract data types from scratch, including binary heap, hash map, linked list, array list, and graph; applied each appropriately to minimize lookup time based on context.
- Implemented an algorithm to find the distance via common movies between 2 distinct cast members.

Resistor Image Scanner

(Python, OpenCV, Flask, HTML, CSS, JS)

A web app that uses image processing techniques to identify resistor values from images; achieved a score of 100%.

- Developed various image transformation pipelines using OpenCV for resistor localisation, image normalisation (denoising, deblurring, removing glare), and final color extraction.
- Implemented the K-Means clustering algorithm to segment color regions and accurately detect resistor band positions.

Discord Drive – Cloud Storage

(Python, Quart, SQLite, HTML, CSS, JS)

A proof of concept full-stack web app allowing users to store files via discord in a server and browse them in a browser.

- Designed and implemented a 3NF SQL database to optimise storage and retrieval of user file data, ensuring efficient and scalable performance.

Gig Management Application

(Java, PostgreSQL, JDBC)

A gig management system to allow venues to seamlessly create, manage and obtain other information about gigs and acts.

- Designed a robust database schema with triggers and views, ensuring robust data validation and providing intuitive database access interfaces.
- Utilised unit testing to validate both the database design and validate the database mutators and accessors, ensuring system reliability.
- Demonstrated proficiency in crafting complex queries to obtain required data from the database through aggregate queries and subqueries.

Rhythm Game Score Showcase Discord Bot

(Python, sqlite)

A Discord bot allowing users to save and showcase scores across multiple servers through messages and custom-generated interactive Discord embeds.

- Implemented asynchronous handling for user interactions to improve system responsiveness.

Other Projects

Rhythm Game Music Player (Java, JavaFX), Maze solver (Java), Basic Circuit Solver (Python Algorithms), Line Following Buggy (C, Memory Mapped I/O), Self Balancing Robot (Hardware Programming, Simulink)